Therefore, we have the following Formula (17).

$$(max_{\mathbf{x}}||\nabla(\sum_{k=1}^{S} g_j^k(\mathbf{x}))||_q) > \sum_{k=1}^{S}(max_{\mathbf{x}}||\nabla(g_j^k(\mathbf{x}))||_q) \tag{17}$$

According to the triangle inequality, we have Formula (18).

$$\begin{aligned} max_{\mathbf{x}}||\nabla(\sum_{k=1}^{S} g_j^k(\mathbf{x}))||_q &\leq max_{\mathbf{x}}(\sum_{k=1}^{S}||\nabla(g_j^k(\mathbf{x}))||_q) \\ &\leq \sum_{k=1}^{S}(max_{\mathbf{x}}||\nabla(g_j^k(\mathbf{x}))||_q) \end{aligned} \tag{18}$$

which contradicts with the derived inequality (Formula (17)). Therefore, the previous assumption does not hold. We show that $\exists\, k, 1 \leq k \leq S, R^k \leq R^{avg}$, demonstrating that the robustness of a member network can be further improved with an ensemble team. Furthermore, for a network F^k, if its robustness bound R^k was not obtained with class j. We have $\exists i \neq j$ $(i, j \neq c)$ and $R^k = \frac{g_i^k(\mathbf{x}_0)}{max_{\mathbf{x}}||\nabla(g_i^k(\mathbf{x}))||_q} \leq \frac{g_j^k(\mathbf{x}_0)}{max_{\mathbf{x}}||\nabla(g_j^k(\mathbf{x}))||_q}$, where Theorem 2 still holds. □

The above analysis formally certifies that an ensemble team of diverse member networks can further improve the robustness of individual networks.

5 EXPERIMENTAL EVALUATION

We conducted a comprehensive experimental evaluation on four benchmark datasets, CIFAR-10, ImageNet, Cora and MNIST, for pruning the given entire ensemble of 10 individual member models for CIFAR-10, ImageNet and Cora and 7 member models for MNIST (see Table 4 for all member models). All the experiments were performed on an Intel i7-10700K server with the NVIDIA GeForce RTX 3090 (24GB) GPU on Ubuntu 20.04.

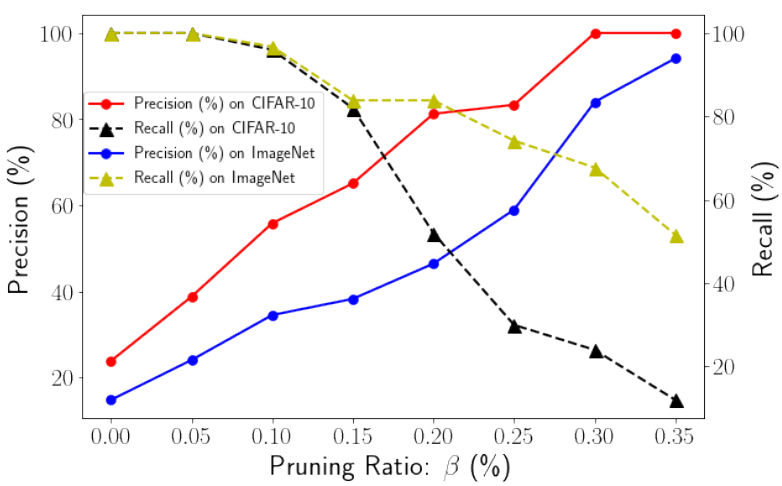

Fig. 4. Impact of β on Precision and Recall (S_d = 4, F-GD)

5.1 Efficiency of Pruning with Varying β

The hyperparameter β in our hierarchical ensemble pruning determines the percentage of candidate ensembles to be pruned out during each iteration for a specific ensemble team size S. In general, a higher β tends to remove more ensemble teams in our hierarchical pruning, potentially lowering the recall score. On the other hand, a high precision score is expected when the ensemble pruning algorithm can precisely find the high quality sub-ensembles with much smaller team sizes and equal or better ensemble accuracy of the reference accuracy of the entire ensemble team. We analyze the impacts of the hyperparameter β in Figure 4, where we vary β from 5% to 35% for both CIFAR-10 and ImageNet using F-GD based hierarchical pruning and the desired team size $S_d = 4$. As shown in Figure 4, when the β increases, the precision of ensemble pruning increases, and the recall decreases for both CIFAR-10 and ImageNet. Our focal diversity based hierarchical ensemble pruning is designed to achieve high precision and good recall, e.g., over 75% precision and 50% recall. Therefore, with $S_d = 4$ on CIFAR-10, we set $\beta = 20\%$ to obtain 81.25% precision and 52% recall of the F-GD based hierarchical pruning. We follow the same principles in determining the β value for other experiments.

Table 8. Impact of Desired Team Size S_d on Hierarchical Pruning (CIFAR-10)

S_d	4	5	6
β (%)	20	10	4
Precision (%)	81.25	**100**	97.56
Recall (%)	52	47.14	57.14

5.2 Impact of Desired Team Size S_d

The hyperparameter S_d specifies the target ensemble team size after ensemble pruning. Table 8 presents the impacts of varying S_d on our focal diversity powered hierarchical pruning approach on CIFAR-10. For each S_d, we find the optimal β following the principles introduced in Section 5.1. We highlight two interesting observations. *First,* for different S_d, our hierarchical pruning can consistently deliver high precision of over 81.25% with good recall, which demonstrates the effectiveness of our hierarchical pruning approach. In particular, $S_d = 5$ produces the highest precision of 100%. *Second,* as the desired team size S_d increases, there is a decrease in the optimal β value, indicating that fewer ensembles will be pruned out for each iteration to achieve a good recall rate. In practice, we recommend setting the desired ensemble team size S_d up to half the size of the entire ensemble, which allows our hierarchical pruning to efficiently find these sub-ensembles that offer comparable or improved ensemble accuracy over the entire ensemble and substantially reduce the ensemble execution cost.

5.3 Focal Diversity Pruning Methods

We then evaluate our hierarchical ensemble pruning algorithm with four focal diversity metrics and the focal diversity consensus voting based refinement. The precision, recall and cost reduction of ensemble pruning are primarily used to measure the pruning efficiency on four benchmark datasets.

CIFAR-10. We compare the four focal diversity metrics using our hierarchical ensemble pruning and the focal diversity consensus voting based pruning (MAJORITY-F) in Table 9, where we set $\beta = 10\%, S_d = 5$, and the F-BD and F-GD pruning results correspond to Figure 3c and 3f. Two interesting observations should be highlighted. *First,* our hierarchical ensemble pruning approach achieves a very high precision of over 85% in ensemble pruning with all four focal diversity

Table 9. Hierarchical Pruning with S_d=5 on CIFAR-10

Methods	F-CK	F-BD	F-KW	F-GD	MAJORITY-F
Precision (%)	85.71	**100**	**100**	**100**	**100**
Recall (%)	17.14	51.43	51.43	52.86	47.14
Cost Reduction	50%	50%	50%	50%	50%

Table 10. Hierarchical Pruning with S_d=4 on CIFAR-10

Methods	F-CK	F-BD	F-KW	F-GD	MAJORITY-F
Precision (%)	26.09	**81.25**	**81.25**	**81.25**	**81.25**
Recall (%)	12.00	52.00	52.00	52.00	52.00
Cost Reduction	60%	60%	60%	60%	60%

metrics. Especially, three focal diversity metrics, F-BD, F-KW and F-GD, produces 100% precision in identifying high quality sub-ensembles with much smaller team sizes and equal or better ensemble accuracy than the entire deep ensemble. *Second,* with the focal diversity consensus voting based ensemble pruning, coined as MAJORITY-F, the 100% precision is maintained. MAJORITY-F further refines the ensemble pruning results by pruning out these ensemble teams that are not chosen by the majority of four focal diversity metrics, which slightly reduces the total number of selected ensembles (i.e., true positives given 100% precision). Hence, the slightly lower recall score of MAJORITY-F is expected in comparison with F-BD, F-KW and F-GD. We find similar observations in Table 10 by setting $\beta = 20\%, S_d = 4$. Even though the F-CK based hierarchical pruning achieves lower performance than the other three focal diversity metrics, the focal diversity consensus voting method, MAJORITY-F, can still deliver consistent high precision of 81.25% and recall of 52%.

Table 11. Hierarchical Pruning with S_d=5 on ImageNet

Methods	F-CK	F-BD	F-KW	F-GD	MAJORITY-F
Precision (%)	0	75.61	75.61	74.42	**78.95**
Recall (%)	0	64.58	64.58	66.67	62.50
Cost Reduction	50%	50%	50%	50%	50%

ImageNet. We then evaluate our hierarchical ensemble pruning approach on ImageNet by setting $\beta = 10\%, S_d = 5$. Table 11 shows the experimental results. We highlight two interesting observations. *First,* our hierarchical ensemble pruning approach performs very well with high precision of over 74.42% by using F-BD, F-KW and F-GD metrics, which is effective and accurate in selecting high quality sub-ensembles with a much smaller team size ($S = 5$ vs. $M = 10$) and higher ensemble accuracy than the 79.82% reference accuracy of the entire ensemble team on ImageNet. On the other hand, the F-CK based pruning failed to identify any satisfactory ensemble teams, which indicates the inherent limitation of the CK diversity, although it has been optimized by the focal diversity measures. *Second,* using our focal diversity consensus voting based ensemble pruning, MAJORITY-F, we can effectively improve the precision by over 3.34% from 74.42%~75.61% to 78.95%, which further demonstrates the enhanced ensemble pruning performance by consolidating the majority of focal diversity metrics.

Cora. We also evaluate and compare our focal diversity based hierarchical pruning approach on Cora, a graph dataset. Table 12 shows the results with $\beta = 10\%$ and $S_d = 4$. All focal diversity based hierarchical pruning methods achieve over 84.93% precision in identifying high quality sub-ensembles with the same or improved ensemble accuracy over 87.90% (the reference accuracy of